

WCD PRACTICAL EXAMINATION

Duration: 60' – Set: 02

Total marks: 100

Requirement:

R1. The details of the Phone of the library are stored in the **Phone** table with the following structure:

Field Name	Type
PhoneID	int identity Primary key
Name	varchar(100)
Price	decimal(15,2)
Image	varchar (255)

- Fill some records for this table [5 marks]

R2. Create a web application for the organization, which can be used by the Admin of the organization for different purpose.

R2.1. login.jsp page: [15 marks]

- For login, the admin should use **Form Authentication** method with an account in **tomcat-users.xml** file or **payara server** (Admin Console). *(Student designs this page yourself)*

R2.2. create.jsp page: [25 marks]

- After login successfully, the **create.jsp** page will be displayed. (5 marks).

Create new Phone

Model:

Price:

Image: No file chosen

- When the “**Create**” button is clicked, the parameters will pass to **PhoneServlet**. (10 marks).
- Validation all fields not blank (5 marks) and price greater than zero (5 marks).

R2.3. PhoneServlet: [35 marks]

- This servlet will be received the parameters from **create.jsp** page (5 marks).
- Process image upload and save to images folder of the web application (10 marks).
- This servlet will use **createPhone(...)** method of the **PhoneBean** to save the information of new Phone to database, use filename of image for image (10 marks).
- If success, the **success.jsp** page will be displayed (5 marks). Otherwise, the **error.jsp** page will be displayed (5 marks).

R2.4. PhoneBean: [20 marks]

- Constructors and Accessor methods (5 marks).
- The **createPhone (...)** method: use **JPA API** to store data in database (15 marks)

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Note: All the marks are evaluated on the GUI